

Roll No.

TBT-701(a)

B. TECH. (BIOTECHNOLOGY)

(SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

ENZYME TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define chromatography. Describe in detail the technique of High Pressure Liquid Chromatography. (CO1)
- (b) Explain in detail some of the important properties of enzymes. (CO1)
- (c) What are proteins ? Explain the process of enzyme isolation from a tissue. (CO1)
2. (a) Explain with the help of a diagram in detail the various features of a substrate velocity plot. (CO2)

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- (b) Explain in detail the following terms : (CO2)
- (i) Enzyme kinetics
 - (ii) Substrate specificity
- (c) Derive mathematically the Michaelis-Menten rate equation. (CO2)
3. (a) What is activation energy ? Explain how enzymes can be used as catalysts to lower the activation energy. (CO3)
- (b) Explain in detail the positive and negative regulation of enzymes. (CO3)
- (c) Elaborate on the competitive, uncompetitive and non-competitive inhibition. (CO3)
4. (a) Explain the role of enzymes in production of alcoholic beverages. (CO4)
- (b) Elaborate of the use of enzymes in the health and pharmaceutical industry. (CO4)
- (c) Discuss the role of enzymes in food industry. (CO4)
5. (a) What is enzyme engineering ? Discuss with applications and advantages. (CO5)
- (b) What is enzyme immobilization ? Discuss its advantages and mention some of the methods for enzyme immobilization. (CO5)
- (c) Explain the gel entrapment method for immobilization in detail. (CO5)

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B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

PHARMACEUTICAL BIOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Elaborate on some of the source of drugs. (CO1)
- (b) Discuss the different routes of drug administration. (CO1)
- (c) Define the importance of parenteral route of drug administration. (CO1)
2. (a) What is difference between pharmacokinetics and pharmacodynamics ? (CO1)
- (b) Elaborate the factor effecting route of drug administration. (CO1)
- (c) Elaborate on the factors that affect the absorption of drug. (CO1)

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3. (a) What is role of drug receptor ? Explain. (CO1)
(b) What is the meaning of bioavailability ? Explain. (CO1)
(c) Elaborate on the mechanism of drug action. (CO1)
4. (a) What is good manufacturing practices ? Explain. (CO2)
(b) What is role of therapeutic proteins in health ? Explain. (CO2)
(c) What is role of biopharmaceutical in treatment ? (CO2)
5. (a) What is Liposome mediated drug delivery and how is it useful in drug delivery ? (CO3)
(b) Write an overview on development of Pharmacology. (CO3)
(c) Write a short note on interleukins and TNF. (CO3)

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B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2022

MICROBIAL TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write short notes on the following : (CO2)
 - (1) Primary and secondary metabolites
 - (2) Single cell protein
 - (3) Recombinant proteins
 - (4) Bioremediation
- (b) Discuss the role of microorganisms in agriculture and pharmaceutical industries. (CO2)
- (c) Discuss the production of recombinant insulin. (CO2)

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2. (a) What do you understand by microbial biomass ? Discuss the steps involved in microbial degradation of cellulose. (CO2 and CO3)
(b) Describe the industrial production of citric acid. (CO2 and CO3)
(c) Implement the knowledge of recombinant DNA technology in production of hGH. (CO2 and CO3)
3. (a) Discuss the Pros and Cons of Genetically Modified Crops. (CO3)
(b) Define the term xenobiotics. How are they harmful for the environment ? Explain with example. (CO3)
(c) Draw the well labelled diagram of Hepatitis B virus and give diagrammatic representation of its vaccine production. (CO3)
4. (a) Discuss the production and applications of xanthan gum. (CO1 and CO4)
(b) Describe industrial production of amoxicillin. Explain its applications also. (CO1 and CO4)
(c) Differentiate between primary and secondary metabolites produced by microbes with examples. (CO1 and CO4)
5. (a) Describe various natural and artificial gene transfer methods. (CO5)
(b) Define vaccine. Explain the types of vaccines with example. (CO5)
(c) Define biopolymers and explain its production. (CO5)

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**B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

STEM CELL TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) How somatic cell therapy can aid in the process of regeneration of the damaged tissue ? (CO3 and CO4)
- (b) Diagrammatically explain the process for the induction of induced pluripotent stem cells and highlight the strategies followed to overcome the limitations of the process. (CO3 and CO4)
- (c) Explain the tests used for the identification of stem cells. (CO3 and CO4)
2. (a) "The set of unspecified cells that have the same developmental potential or ability to idopt various fates." Comment on the aforementioned statement. (CO1 and CO2)

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- (b) "The process which helps in the neuron development." Explain such process with the help of a diagram. (CO1 and CO2)
- (c) Describe the process of DNA methylation and give its significance. (CO1 and CO2)
3. (a) Differentiate between In-vitro fertilization and somatic cell nuclear transfer. (CO1 and CO5)
- (b) "Pathways that aid in the cell fate determination". Explain any *one* pathway relevant to the statement mentioned. (CO1 and CO5)
- (c) How embryonic stem cell can help in replacing destroyed brain cells with healthy, undifferentiated stem cells? (CO1 and CO5)
4. (a) Explain the types of cell assays which has been considered important for the identification of stem cell's ability in transplantation. (CO5 and CO1)
- (b) How stem cell could be used as a weapon for an efficient drug delivery process? (CO5 and CO1)
- (c) Diagrammatically explain the methodology adopted for the isolation of stem cells. (CO5 and CO1)
5. (a) Discuss the methods adopted for the storage of stem cells. Also highlight the contamination caused during storage. (CO3 and CO5)
- (b) Differentiate between tissue engineering and genetic engineering. (CO3 and CO5)
- (c) How do epidermal stem cells regulate the process of hair folliculogenesis? (CO3 and CO5)

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B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2022

MARINE BIOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Elaborate on some of the economically important finfishes in India. (CO1)
- (b) Discuss some of the economically important oysters in India. (CO1)
- (c) Explain in detail some of the economically important crabs and shrimps in India. (CO1)
2. (a) What are extremophiles ? Write short notes on any *three* types of extremophiles giving an example of each. (CO2)
- (b) What are non-culturable microbes ? Elaborate on the role of microbes in a marine ecosystem. (CO2)

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- (c) Elaborate on the technique of metagenomics and explain its role in studying *e*-DNA. (CO2)
3. (a) Explain in detail some of the important bioactive molecules extracted from marine sources. (CO3)
- (b) What are marine toxins ? Elaborate on the use of some of the marine toxins in treating diseases. (CO3)
- (c) Write in detail on the following marine products : (CO3)
- (i) Tetrodotoxin
- (ii) Omega 3 Fatty acid
4. (a) What is Bioremediation ? Explain how it can be used to restore marine ecosystems. (CO4)
- (b) What is algal bloom ? Explain what are the disadvantages of algal bloom for marine ecosystem and how can it be prevented. (CO4)
- (c) Elaborate on the methods of degradation of pesticides and heavy metals in marine ecosystem. (CO4)
5. (a) Elaborate on some of the diseases associated with cultured shrimps. (CO5)
- (b) Explain the role of biosensors in marine environment giving appropriate examples. (CO5)
- (c) Discuss how molecular tools be used to diagnose diseases of marine organisms. (CO5)

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B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

FOOD BIOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain various historical background events in food biotechnology.

(CO1)

(b) Discuss Positive and Negative interactions of microorganisms with food.

(CO1)

(c) Explain how temperature, water activity, pH, gases concentration and physical characteristics of food affect microorganisms interaction with food.

(CO1)

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2. (a) Discuss fermentation of dairy products in detail. (CO2)
(b) Discuss process of malting. Enlist various uses of malt. (CO2)
(c) Discuss the term SCP, its different sources and its importance. (CO2)
3. (a) Discuss various methods utilized for enumeration and detection of food-borne organisms. (CO3)
(b) Write short note on emerging preservation techniques for milk and dairy products. (CO3)
(c) Discuss role of various microorganisms involved in spoilage of milk and milk products. (CO3)
4. (a) Discuss the effect of irradiation on food constituents. (CO4)
(b) Discuss classification of irradiation on the basis of dosage utilized and their purpose. (CO4)
(c) Discuss the working mechanisms of food irradiation, its uses, advantages and legal status. (CO4)
5. (a) (i) Differentiate between chilling and freezing.
(ii) Differentiate between slow freezing and fast freezing. (CO5)
(b) Discuss, symptoms, food responsible and their precaution for following food-borne infection : (CO5)
(i) Salmonellosis
(ii) Clostridium perfringens gastroenteritis
(iii) Botulism poisoning
(c) Discuss different methods in prevention of food by high temperature, their advantages and drawbacks. (CO5)

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**B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

IPR, BIOSAFETY AND BIOETHICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain utility of biotechnology inventions. (CO1)
(b) Describe issues concerning biopiracy. (CO1)
(c) What is *sui generis* system ? Why is *sui generis* system important ? (CO1)
2. (a) Give complete procedure of the filing of patent. (CO2)
(b) Give a detailed account of GATT agreement stating its pros and cons. (CO2)
(c) Describe trademark registration process and trademark infringement. (CO2)

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3. (a) Explain concept of biosafety and biorisk. (CO3)
(b) Explain hazardous characteristics of the agents and good lab practices. (CO3)
(c) Explain biosafety levels to personal, environment and community. (CO3)
4. (a) Explain principles of Lab biosecurity. (CO4)
(b) Describe transportation of infectious substances. (CO4)
(c) What are the major components of integrated pest management with guidelines ? (CO4)
5. (a) Comment on the significance of Bioethics. (CO5)
(b) Explain future bioethical conflicts in biotechnology. (CO5)
(c) Describe resolution of conflicts. (CO5)